

SMART INDIA HACKATHON 2026



SETU · INTELLIGENT DEAD RECKONING

Problem Statement ID – _____

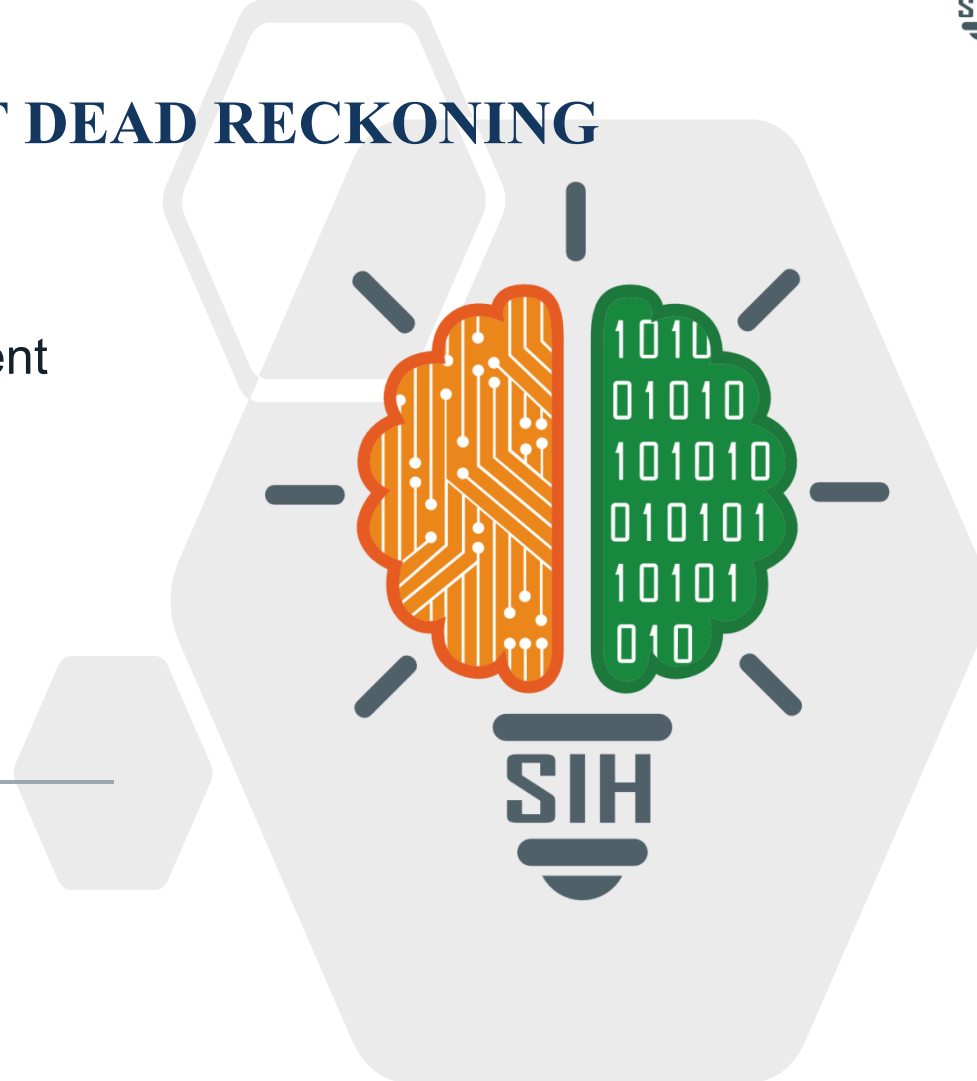
Problem Statement Title- AI-ML based Intelligent Dead Reckoning system for seamless navigation

Theme- _____

PS Category- Software

Team ID- _____

Team Name (Registered on portal)- _____



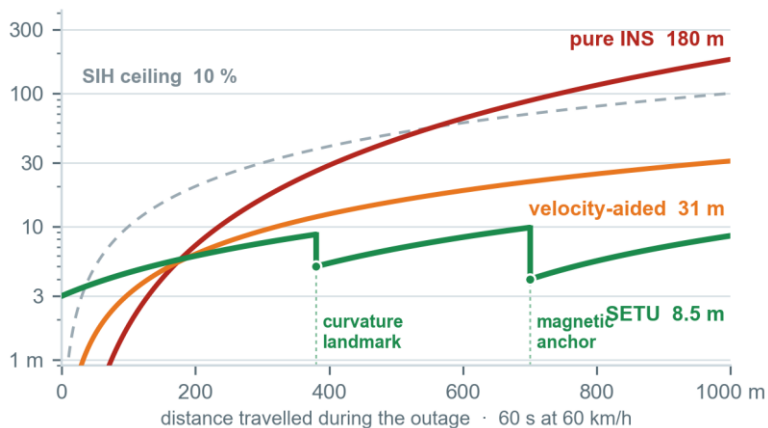
Every phone-only system integrates acceleration twice, so its error grows as t^2 . SETU asks a different question — how far along this road am I — and answers it with three measurements whose error does not accumulate with time.

Detailed explanation of the proposed solution

- One C++ engine, two shells: Android app and edge daemon.
- ACE** solves phone→vehicle pitch, roll and yaw online. No user calibration; re-aligns in under 2 s after a knock.
- MSVR** classifies motion and mount quality, so potholes, idling and phone handling gate every other module.
- SVO** reads ground speed off the axle harmonics already present in the accelerometer spectrum.
- CTS** reads absolute speed from lateral force ÷ yaw rate in every turn — lean-corrected for two-wheelers.
- CSA** registers the gyro-derived path shape onto an offline OSM road manifold.
- EFA** magnetic, barometric and radio anchors, built by ordinary driving rather than by a survey fleet.
- IAF** invariant EKF on $SE_2(3)$ with learned noise, inside a road-graph particle filter.
- GQM** per-satellite trust from raw GNSS — handover ramps, never switches.

3.6 MB int8 models 2.7 ms per tick 10 Hz phone 200 Hz edge
 · fully offline

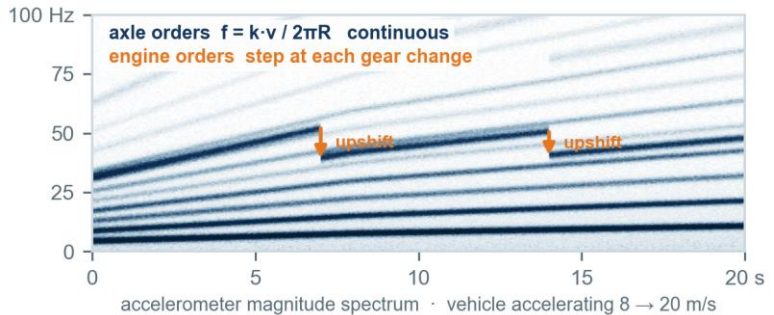
How it addresses the problem



- Integrating** a 0.1 m/s² bias gives $\frac{1}{2} \cdot b \cdot t^2 = 180$ m in one minute. No network repairs a double integrator.
- Aiding velocity** makes the error linear in distance — 31 m per km at the 3.1 %/km published for a smartphone in a real tunnel.
- Registering position** makes it bounded: error resets at every curvature landmark and field anchor, so it stops growing with time.

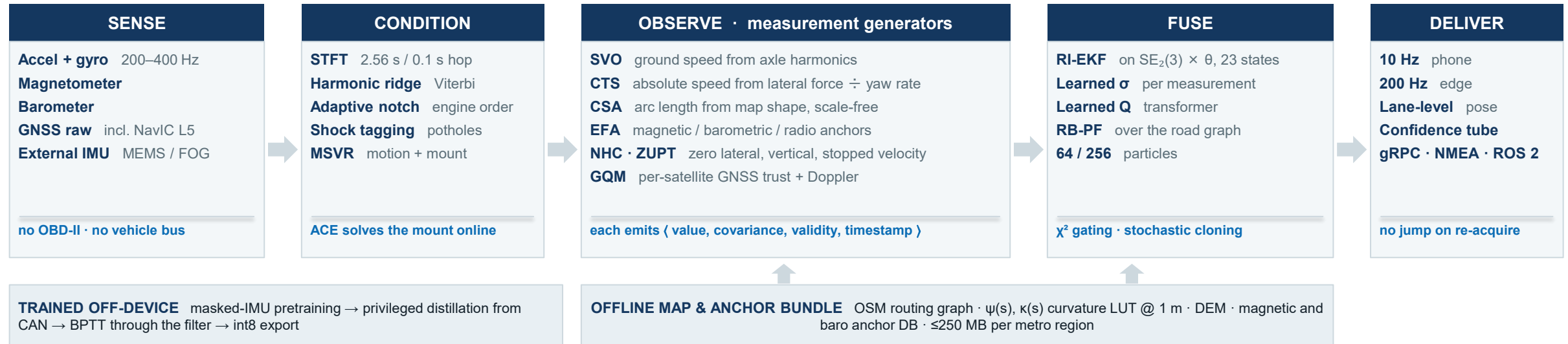
Innovation and uniqueness of the solution

- N1** Speed from frequency, not integration. Chassis-vibration speed tracking exists in the DSP literature and has never been fused into a navigation filter.
- N2** Two-wheeler physics done right: $v = g \cdot \sin\phi / \omega z$. Omitting $\cos\phi$ is a silent 4 % scale error — 40 m per km.
- N3** Scale-free map registration. Prior shape-matching work needs an odometer; we solve for the scale itself.
- N4** IO-VNBD's synchronised CAN wheel speeds used as a training-only teacher instead of noisy GPS speed.
- N5** Learned σ and learned Q inside one invariant filter, multi-hypothesis over the road graph.

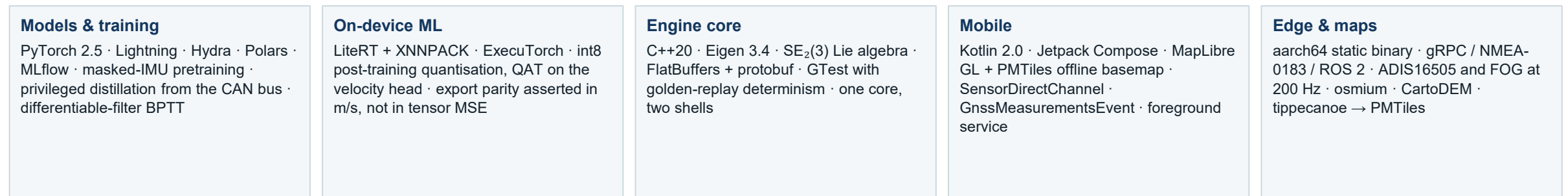


N1 evidence — axle orders rise continuously with speed while engine orders step down at each upshift. That step is a free training label.

Methodology and process for implementation



Technologies to be used



Analysis of the feasibility of the idea

Data already in hand. IO-VNBD gives 5,700 km over 98 h with the phone IMU and the CAN bus recorded simultaneously — both the input and a clean label.

The load-bearing physics is closed-form. $v = a_{lat} / \Omega$ and $f = v / 2\pi R$ are algebra, not training. Only the residuals are learned, so the core carries no research risk.

Compute fits today's phones. 3.6 MB of int8 models, 2.7 ms per 10 Hz tick, CPU only — no NPU or GPU delegate dependency across low-end SKUs.

The sensors are already there. Accelerometer, gyroscope, magnetometer, barometer and raw GNSS including NavIC L5 are exposed by standard Android APIs.

Maps are tractable offline. ≤ 250 MB per metro region for the routing graph, the precomputed curvature lookup and the DEM. No connectivity in the runtime path.

Nothing to buy or install. No OBD-II dongle, no wheel encoder, no UWB, leaky-feeder or 5G roadside build-out — it runs on the phone already on the dashboard.

Capability tiers — declared at runtime, never hidden

A

≥ 200 Hz IMU full spectral odometer

0.5–1 %/km

B

50–200 Hz axle orders 1–4 only

1–2 %/km

C

10–50 Hz no spectral odometer

2–3 %/km

IO-VNBD's smartphone stream is 10 Hz, so every result we report on it is Tier C with the spectral odometer switched off.

Potential challenges and risks	Severity	Strategies for overcoming these challenges
CSA is ambiguous on featureless or badly-mapped roads	High	Landmark saliency scoring, top-k path hypotheses in the particle filter, honest σ from the Gauss–Newton Hessian. Fallback: classical HMM map matching, still inside the gate.
Axle harmonics too weak — smooth road, quiet EV, soft mount	Medium	Mount-quality gate disables SVO when the mechanical path is decoupled; summation over 8+ orders; learned σ down-weights it automatically. Fallback: Tier B, gate still met.
IO-VNBD's phone stream is 10 Hz, so it cannot validate SVO	Known limit	Declared as capability tiers rather than hidden. IO-VNBD results are reported as Tier C with SVO off; SVO is validated on self-collected 400 Hz logs from week 2.
No two-wheeler data in IO-VNBD — four cars, no Indian roads	Known limit	The lean relation is imposed as a physics loss rather than learned, so few samples are needed; a dedicated two-wheeler collection covers both mount types.
Phone sensor rate throttled below 200 Hz on some devices	Medium	Tier detected at runtime from the achieved rate; SensorDirectChannel where available; verified across high, mid and low device classes.
Uneven OpenStreetMap geometry on Indian roads	Medium	Cross-track covariance scaled by an OSM confidence heuristic, an explicit off-road particle so a missing road cannot force a wrong match, and a provider adapter.

Potential impact on the target audience

- Quick-commerce and delivery riders** — the marker keeps moving through underpasses and basement parking, so the next turn is announced before the exit, not after it.
- Logistics and fleet operators** — mileage through tunnels and yards stops being guesswork; ETA and proof of delivery survive GNSS gaps, with no dongle to fit or maintain.
- Ride-hailing drivers and riders** — no frozen or teleporting car in an urban canyon, and pickup points that resolve to the right side of the road.
- Emergency responders** — ambulance and fire crews keep a live position inside tunnels, metro underpasses and multi-level structures, exactly where dispatch loses them today.
- Two-wheeler riders** — the largest vehicle class on Indian roads gets a lean-aware solution instead of one built for cars, and less time spent looking at a stalled screen.
- Defence and critical logistics** — a wholly self-contained position that degrades gracefully under jamming or spoofing instead of failing.

Where it matters most

Long tunnels and metro underpasses — GNSS is absent, and the road is topologically one-dimensional, so the map does most of the work.

Multi-level parking — the helical ramp is a uniquely-shaped signature and the barometer resolves the floor to within one level.

Dense urban canyons — position is unusable but Doppler velocity survives on three or four satellites, where others discard everything.

Forested and valley highways — partial, intermittent GNSS, which the trust ramp absorbs without ever switching mode.

Benefits of the solution (social, economic, environmental, etc.)

- Social**

Safer riding: a correct instruction on time means fewer glances at the screen at the moment a rider can least afford one. Emergency crews stay locatable underground. Accurate navigation stops being a feature of expensive cars and becomes a property of any phone on any dashboard.

Economic

Zero marginal hardware — it runs on phones riders already own. No OBD dongle per vehicle, no factory-fitted INS, no UWB or leaky-feeder installation per tunnel. Fewer missed exits, re-routes and failed delivery attempts per shift, on the same fleet.

Environmental

A missed exit is wasted kilometres, wasted fuel and avoidable emissions; removing the cause removes all three. And because the solution is entirely on-device, no roadside infrastructure has to be manufactured, powered or maintained.

Strategic

Resilient to jamming and spoofing by construction, since every SETU channel is self-contained. Uses NavIC L5 where Indian devices expose it, runs fully offline, and keeps trajectory data on the phone rather than in a cloud.

Pure inertial, 60 s

180 m

$\frac{1}{2} \cdot b \cdot t^2$ with a 0.1 m/s² bias — the marker is a block away

Best published smartphone

31 m

3.1 % of distance per km, INS + NHC in a real tunnel

SETU design target

8–15 m

0.8–1.5 % per km, anchored to the map and to the field

Horizontal drift over a 1 km GNSS blackout at 60 km/h. The SIH ceiling is 10 % of distance travelled — 100 m.



RESEARCH AND REFERENCES



Details / Links of the reference and research work

DATASET AND BENCHMARK

IO-VNBD: Inertial & Odometry Benchmark Dataset

Onyekpe, Palade, Kanarachos, Szkolnik · Data in Brief 35:106885, 2021 · github.com/onyekpeu/IO-VNBD · [arXiv:2005.01701](https://arxiv.org/abs/2005.01701)

WhOnet: Wheel Odometry Neural Network

Eng. Applications of AI, 2021 · [arXiv:2104.02581](https://arxiv.org/abs/2104.02581) — our privileged upper bound

R-WhOnet: recalibration via transfer learning

[arXiv:2209.05877](https://arxiv.org/abs/2209.05877) — evidence for cross-vehicle domain shift

LEARNED INERTIAL ODOMETRY

Deep Learning for Inertial Positioning: A Survey

Chen & Pan · IEEE T-ITS, 2024 · [arXiv:2303.03757](https://arxiv.org/abs/2303.03757)

Inertial Navigation Meets Deep Learning — a survey

Cohen & Klein · 2024 · [arXiv:2307.00014](https://arxiv.org/abs/2307.00014)

AI-IMU Dead-Reckoning — invariant EKF, learned noise

Brossard, Barrau & Bonnabel · IEEE T-IV, 2020 · [arXiv:1904.06064](https://arxiv.org/abs/1904.06064)

TLIO: Tight Learned Inertial Odometry

Liu et al. · IEEE RA-L, 2020 · [arXiv:2007.01867](https://arxiv.org/abs/2007.01867) — learned covariance into a filter

OdoNet: untethered speed aiding, no odometer

[arXiv:2109.03091](https://arxiv.org/abs/2109.03091) · 68 % error reduction over NHC alone

CarSpeedNet: accelerometer-only speed

[arXiv:2401.07468](https://arxiv.org/abs/2401.07468) · <0.72 m/s $\approx$ 4.3 % — the accuracy to beat

PiDR: Physics-Informed Inertial Dead Reckoning

Sahoo & Klein · 2026 · [arXiv:2601.03040](https://arxiv.org/abs/2601.03040)

LEARNED AND ADAPTIVE FUSION

KalmanNet: neural-network-aided Kalman filtering

Revach, Shlezinger, van Sloun & Eldar · IEEE TSP, 2022

A-KIT: Adaptive Kalman-Informed Transformer

Cohen & Klein · [arXiv:2401.09987](https://arxiv.org/abs/2401.09987) · github.com/ansfl/A-KIT — learned process noise

Differentiable particle filters, semi-supervised

[arXiv:2011.05748](https://arxiv.org/abs/2011.05748)

MAP MATCHING AND KINEMATIC CONSTRAINTS

Hidden Markov Map Matching Through Noise and Sparseness

Newson & Krumm · ACM SIGSPATIAL, 2009 — the map-matching baseline

Heading-length sequence matching for localisation

[arXiv:2005.13704](https://arxiv.org/abs/2005.13704) — nearest prior art to CSA; assumes a known speed source

Map-aided dead reckoning from OBD speed

[arXiv:1611.07910](https://arxiv.org/abs/1611.07910) — map-aided DR with a known odometer

Lever-arm accuracy of the non-holonomic constraint

Zhang & Hu · IEEE TVT, 2020

NHC-assisted GNSS/SINS with motion-state CNN

GPS Solutions, 2023 · doi 10.1007/s10291-023-01483-9

SPECTRAL SPEED AND FIELD ANCHORS

Vehicle Speed Tracking Using Chassis Vibrations

Linköping University, 2016 — axle-order tracking; motivates tunnel navigation

Accelerometer-based wheel odometer

Sensors, 2021 · PMC7918720

Magnetic field vehicle positioning in a long tunnel

Applied Sciences 11(24):11641, 2021 · doi 10.3390/app112411641

Vehicle positioning in tunnel environments: a review

Complex & Intelligent Systems, 2025 · doi 10.1007/s40747-024-01744-1

Barometric floor detection on smartphones

Sensors, 2015 · PMC4431287 — basis for the parking-level anchor

GNSS INTEGRITY AND PLATFORM

Vehicle positioning in underground space using a smartphone

ISPRS Archives XLVI-3/W1, 2022 — the 3.1 %/km INS+NHC tunnel reference

GNSS multipath detection using machine learning

Hsu · IEEE ITSC, 2017

Android GnsMeasurement HAL — raw GNSS observables

source.android.com/reference/hal/struct_gnss_measurement

u-blox Untethered Dead Reckoning (UDR)

u-blox.com/en/technologies/udr-untethered-dead-reckoning — commercial reference

A 66-entry annotated bibliography, with a prior-art positioning table for each of the five mechanisms, is maintained alongside the implementation documentation.